

SGA 2: Application to the Fundamental Group

Bounded source-aligned working unit — 20 July 2026

Authority. Corrected French TeX lines 3414–3423; original printed p. 116, physical source-PDF pp. 100–101, and recomposed running pp. 92–93. These locator systems are kept distinct. Blank line 3424 is excluded; the raw cursor is line 3424 and the next substantive cursor is line 3425 (the proof). The complete proposition, parallel module/finite-flat/étale cases, symbols, and page layout were checked directly. The external English candidate is comparison-only.

Proposition 2.3. Let X be a locally noetherian prescheme, and let Y be a closed subset of X . Let $\widehat{X}$ be the formal completion of X along Y . For every open subset U of X containing Y , let L_U (respectively P_U , respectively E_U) denote the functor that assigns to every locally free coherent $\mathcal{O}_U$ -module (respectively every finite flat covering of U , respectively every étale covering of U) its pullback along $\widehat{X} \rightarrow X$.

- (i) If $\text{Lef}(X, Y)$ holds, then, for every open neighborhood U of Y , the functors L_U , P_U , and E_U are fully faithful.
- (ii) If $\text{Leff}(X, Y)$ holds, then, for every locally free coherent $\mathcal{O}_{\widehat{X}}$ -module $\mathcal{E}$, there are an open neighborhood U of Y and a locally free coherent $\mathcal{O}_U$ -module E such that

$$L_U(E) \simeq \mathcal{E}.$$

The analogous assertions hold for finite flat coverings and for étale coverings, with P_U and E_U , respectively.